Exploring Excess Return Generation through European Fixed Income Arbitrages

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Abstract. We build systematic trading strategies on three European fixed-income arbitrages that the literature has documented as pricing anomalies but never traded: the BTP Italia inflation-linked basis, the corporate CDS–bond basis, and the iTraxx index skew. Net of regime-dependent transaction costs the three earn annualized Sharpe ratios of 0.98 to 1.56 with positive skewness and drawdowns below 4.5%. Alpha of 2.2% to 4.8% per year survives three canonical factor benchmarks, eight principal components, and a best-subset search over 75 tradable factors, in sample and out; model-X knockoffs at a 10% false discovery rate retain no factor at all. The premium roughly triples between calm and stressed markets. Our contribution is to turn the cross-section into a test of *why* it survives. The three trades are run by the same leveraged arbitrageurs but differ in who holds the underlying instrument: two sit on institutional balance sheets, the BTP Italia sits with unlevered households. A single funding shock therefore widens the two institutional mispricings together while pulling the retail-held one the other way—a sign structure no common risk factor can generate, and direct evidence that slow-moving intermediary capital, not risk, is what these strategies are paid for. For investors this is a state-dependent premium with an internal hedge, kept open by post-crisis regulation rather than by mispricing that is about to close.

1 Introduction

Duarte et al. (2007) ask whether fixed-income arbitrage earns genuine alpha or merely picks up "nickels in front of a steamroller." We take their question to European markets and to three strategies that have each been documented as a pricing anomaly but never examined as a systematic trading strategy: the inflation-linked basis on Italian BTP Italia bonds, the European corporate CDS–bond basis, and the iTraxx index skew—the gap between the traded index spread and the spread implied by its single-name constituents.

Three features make these trades unusually clean laboratory material. First, each has a *deterministic payoff at maturity*: held to maturity the profit is locked in regardless of the interim path. The strategies in Duarte et al. (2007)—swap spreads, curve trades, mortgage hedging, volatility selling—do not share this property; their payoffs depend on prepayment models, mean-reversion assumptions, or realized volatility, so the outcome is uncertain even at maturity. Second, all three are implementable: entry and exit follow mechanical basis thresholds, transaction costs are charged at the trade level and widen with market stress, and every return we report is net of those costs. Third, and central to this paper, the three trades are run by the same type of leveraged institutional arbitrageur but differ in *who holds the underlying instrument*. The CDS–bond and iTraxx legs sit on institutional balance sheets. The BTP Italia, by design of the Italian Treasury, is a retail instrument held by unlevered households.

That last asymmetry is what converts a performance study into an identification design. The slow-moving capital literature—Duffie (2010), Brunnermeier and Pedersen (2009), Mitchell and Pulvino (2012)—holds that mispricings persist because the capital that would close them is scarce and arrives slowly, and that mispricings drawing on a *common pool* of intermediary capital must therefore co-move, and co-move harder when the pool is depleted. The competing explanation is that the three bases simply load on a common macroeconomic risk factor. The two hypotheses are observationally similar in a conventional cross-section, because both predict co-movement. They separate here. Under the capital channel, co-movement should depend on whether the holder base is shared, so the retail-held basis should sit outside the pool—and, if forced institutional selling coincides with household demand for the linker, should move the *other way*. Under a macroeconomic factor, all three bases move alike, whoever holds them. The sign is the test.

We report three results. (i) All three strategies earn positive, economically meaningful excess returns net of costs, with *positive* return skewness—the opposite of the steamroller caricature. (ii) The alpha is not compensation for identified systematic risk: it survives five progressively harder layers of factor controls, including a supervised search that is free to construct the most absorbing hedge available from 75 tradable candidates, and it survives out of sample with hedge loadings estimated on past-only windows. (iii) The premium concentrates in intermediary funding stress, and the cross-sectional behavior of the mispricings carries the signature of the capital channel: the two institutionally held bases co-move ever more tightly as capital tightens, while the retail-held BTP Italia moves against them.

Relative to the literature the contribution is threefold. The BTP Italia basis has, to our knowledge, never been studied; its persistence is the sharper puzzle because the instrument eliminates the ballooning risk that complicates the TIPS–Treasury basis of Fleckenstein et al. (2014), so the two legs of the trade carry identical sovereign credit exposure. Neither the euro-area negative basis trade nor the iTraxx skew has been examined

Table 1: Performance of the three arbitrage strategies, net of transaction costs. Monthly signal-weighted excess returns; mean and volatility annualized; t-statistics use Newey–West HAC standard errors with the Newey and West (1994) truncation lag ($m = 4$ for every sample). MaxDD is the deepest peak-to-trough drawdown of the cumulative index. MPPM is the manipulation-proof performance measure of Goetzmann et al. (2007) at risk aversion $\rho = 3$, which cannot be inflated by information-free dynamic trading or leverage—the appropriate yardstick for threshold-driven strategies. All samples end in May 2025.

Strategy	n	Mean (% p.a.)	t	Vol. (% p.a.)	Sharpe	Skew	MaxDD (%)	MPPM (% p.a.)
BTP Italia (retail-held)	163	1.59	3.52	1.62	0.98	+0.81	−3.0	1.48
CDS–Bond Basis	207	4.61	5.87	2.96	1.56	+0.79	−4.4	4.39
iTraxx Skew	243	1.91	4.92	1.53	1.25	+1.01	−2.5	1.87

as a systematic strategy—Junge and Trolle (2015) and Boyarchenko et al. (2016) study the skew as a gauge of arbitrage limits, not as a tradable position. And where the existing work tests slow-moving capital on markets that share a holder base, the European cross-section lets us vary the holder base while holding the arbitrageur fixed.

2 Three European Fixed-Income Arbitrages

Each arbitrage is defined by a basis that no-arbitrage forces toward a well-defined level. Following Duarte et al. (2007), each trade is modelled as a separate fictional fund that opens when the basis crosses an entry threshold and closes on reversion, on maturity, or at the end of the sample. Thresholds are in Appendix A.

BTP Italia. The BTP Italia is indexed to Italian CPI and differs from conventional linkers in two ways that matter here. It pays the inflation revaluation with each semiannual coupon and redeems at par, so the notional exposed to credit risk never balloons and the two legs of the basis trade carry identical sovereign exposure; and it is placed primarily with households, through a retail-only auction phase and a loyalty bonus (*premio fedeltà*) for buy-and-hold investors. The basis is the fixed-rate-equivalent yield of the BTP Italia—converted with the term structure of zero-coupon inflation swaps, following the replication logic of Fleckenstein et al. (2014)—minus the yield of the maturity-matched nominal BTP. Both legs are general collateral in the euro repo market, so funding-cost differentials cannot explain the gap.

CDS–bond basis. For each issuer in the on-the-run iTraxx Main universe the basis is the CDS spread minus the bond z-spread. When it is negative, buying the bond and buying protection locks in the gap and returns par in default, since the CDS tenor is matched to the bond's residual maturity. We trade the negative basis only: the reverse trade requires a corporate reverse repo subject to recall risk precisely in stress (Bai and Collin-Dufresne 2019). This is the one trade of the three that must finance a cash bond on a dealer balance sheet—a fact we exploit in Section 4.

iTraxx skew. The skew is the traded index spread minus the intrinsic spread bootstrapped from the constituents' CDS curves. We trade it across the four iTraxx sub-indices (Main, SnrFin, SubFin, Xover), on-the-run only, with both legs under a single netting set. It is a derivative-on-derivative trade with no cash leg to finance.

Individual trades aggregate into signal-weighted indices, in which weight is proportional to the entry dislocation from the expanding-window historical equilibrium, following the curvature-signal logic of Rebonato and Ronzani (2021); equal-weighted results, which are close, are in Appendix A. Transaction costs are charged per trade in proportion to DV01, at fee tiers that rise with the level of the iTraxx Main 5-year spread—LOW below 60 bps, MEDIUM to 100 bps, HIGH above—so costs are most conservative exactly when gross arbitrage returns are largest. The same three regimes classify market conditions throughout the paper.

Table 1 reports the record. The CDS–bond basis is the strongest performer on every measure; the iTraxx skew has the longest sample and the BTP Italia the shortest, beginning with the instrument's first issuance in 2012. Two features deserve emphasis. First, all three return distributions are *positively* skewed, which cuts directly against the characterization of fixed-income arbitrage as selling disaster insurance, and the deepest drawdown of the three is 4.4%, recovered within months. Second, the MPPM is essentially flat in risk aversion—1.50, 1.48, 1.47 for BTP Italia across $\rho \in \{2, 3, 4\}$, and similarly for the others (Appendix A)—so an investor with $\rho = 4$ values these strategies almost identically to one with $\rho = 2$. Risk-adjusted performance

Table 2: Annualized alpha (% p.a.) across five layers of factor controls. Panel A: full-sample spanning regressions, Newey–West t-statistics, adjusted R^2 in brackets. Panel B: out-of-sample alpha in the recursive design of Welch and Goyal (2008)—at each month the hedge loadings are re-estimated on a past-only expanding window (60-month burn-in) and alpha is the mean realized abnormal return $r_{i,t} - \hat{\beta}'_{i,t-1} f_t$, with the in-window intercept excluded from the hedge. For the PCA the components are re-extracted inside each window, so the design is invariant to PCA sign and rotation indeterminacy; for the best subset the factor set is frozen and only loadings are re-estimated. ***,**,*: 1%, 5%, 10%. Every best-subset out-of-sample t-statistic clears the $t > 3.0$ hurdle Harvey et al. (2016) propose for discoveries in multiple-testing environments.

Control layer	BTP Italia	CDS–Bond	iTraxx Skew
Panel A: In-sample alpha $[\bar{R}^2]$			
Duarte et al. (2007), $k = 10$	$+1.6^{***}$ [0.11]	$+4.2^{***}$ [0.15]	$+2.2^{***}$ [0.06]
Brooks et al. (2020), $k = 8$	$+2.1^{***}$ [0.08]	$+4.2^{***}$ [0.12]	$+1.7^{***}$ [0.04]
Fung and Hsieh (2004), $k = 7$	$+1.8^{***}$ [0.10]	$+4.0^{***}$ [0.11]	$+1.8^{***}$ [0.00]
Principal components, $K = 8$	$+1.4^{***}$ [0.09]	$+4.8^{***}$ [0.11]	$+2.2^{***}$ [0.13]
Best subset (ℓ_0), $k = 8$	$+2.2^{***}$ [0.31]	$+4.8^{***}$ [0.21]	$+2.9^{***}$ [0.25]
Panel B: Out-of-sample alpha (expanding window)			
Duarte et al. (2007)	$+1.29^{**}$	$+3.48^{***}$	$+2.27^{***}$
Brooks et al. (2020)	$+2.10^{***}$	$+3.86^{***}$	$+1.77^{***}$
Fung and Hsieh (2004)	$+1.69^{***}$	$+3.45^{***}$	$+2.02^{***}$
Principal components	$+1.61^{***}$	$+4.55^{***}$	$+1.85^{***}$
Best subset (frozen set)	$+2.05^{***}$	$+3.90^{***}$	$+2.50^{***}$

does not rest on tail events. A Moreira and Muir (2017) volatility-managed overlay adds nothing for the two unfunded trades (-0.22% and -0.16% p.a., both insignificant) and a marginally significant $+1.39\%$ for the CDS–bond basis—the first hint that the one trade that must be financed behaves differently from the two that need not, a distinction Section 4 turns into a test.

3 Is It Alpha? Five Layers of Risk Controls

Sharpe ratios are not alpha. We subject the three return series to five progressively harder spanning regressions $r_{i,t} = \alpha_i + \beta_i' f_t + \varepsilon_{i,t}$, in which every regressor is a realized excess return, so each intercept keeps its interpretation as a Jensen alpha. The first three are fixed benchmarks from the fixed-income and hedge-fund literatures: Duarte et al. (2007) ($k = 10$ equity, banking and term-structure factors), Brooks et al. (2020) ($k = 8$ active fixed-income risk premia), and Fung and Hsieh (2004) ($k = 7$ asset-based style factors including trend-following lookback straddles). Two ingredients of the originals are replaced with tradable counterparts—yield and spread *changes* become total-return series—so the intercept remains an implementable abnormal return. Because all three strategies trade European instruments, the euro panel is the primary specification; US-factor estimates are in Appendix B.

Fixed benchmarks can only absorb the exposures their authors chose to include. The last two layers let the data choose. The fourth extracts the first $K = 8$ principal components of a panel of 75 tradable candidate factors—credit, liquidity and funding, equity, volatility, and rates—selected on a published link to arbitrage returns or intermediary constraints, with K set by the Bai and Ng (2002) IC_{p2} criterion (64.7% of panel variance) and every state variable entering through its traded counterpart, as the He et al. (2017) intermediary capital factor does (Connor and Korajczyk 1986). The fifth is best-subset selection (Bertsimas et al. 2016): for each strategy an ℓ_0-constrained search picks whichever eight of the 69 de-duplicated candidates explain its returns best. This is the hardest test we can pose, because the data are free to construct the most absorbing hedge in the universe.

Table 2 reports the outcome, and the shape of the table is the result. Reading down Panel A, explanatory power rises from near zero to adjusted R^2 of 21–31%—the supervised hedge does find real exposures—while the alpha barely moves: 1.4–2.2% per year for BTP Italia, 4.0–4.8% for the CDS–bond basis, 1.7–2.9% for the iTraxx skew, significant at the 1% level in every cell. A factor structure that explains a third of the return variance absorbs none of the mean. Panel B repeats the exercise in real time, and the alpha survives everywhere.

Three details sharpen the verdict. First, the three selected factor sets are pairwise *disjoint*—no factor appears in more than one strategy—so no single common exposure is responsible, and the selected loadings

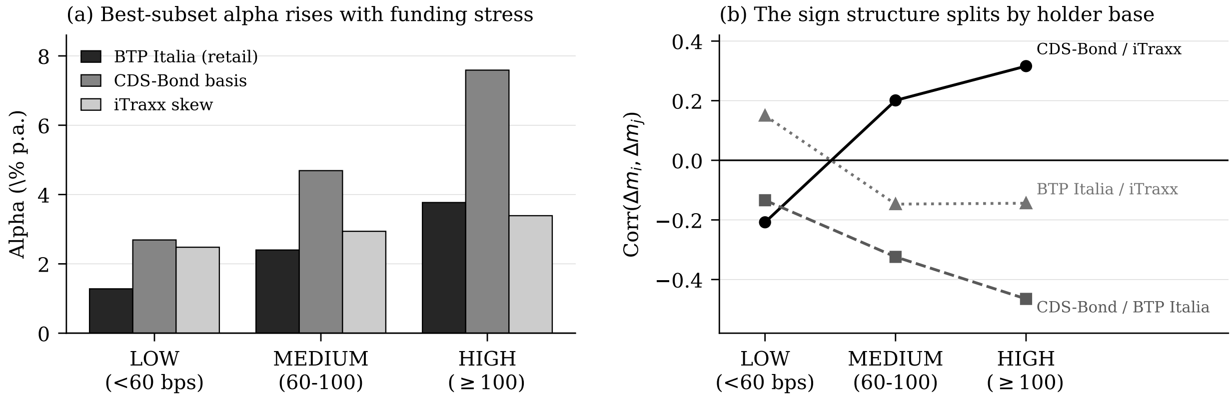

Figure 1: The premium and the sign structure, by funding-stress regime. Regimes on the 5-year iTraxx Europe Main spread (LOW < 60, MEDIUM $[60, 100)$, HIGH ≥ 100 bps). Panel (a): annualized alpha per regime from a single regression with three intercept dummies and betas common across regimes, on the best-subset factors. Panel (b): within-regime Pearson correlations of monthly mispricing changes Δm. The institutional pair (solid) rises monotonically through zero; both pairs that cross holder bases fall. Full estimates in Appendix D.

map onto each trade's structure: BTP Italia loads on inflation, liquidity, and sovereign credit; the CDS–bond basis on the prime-broker CDS factor and the default premium; the iTraxx skew on the traded He et al. (2017) intermediary capital factor ($t = -4.1$), meaning it pays off when dealer equity underperforms. Second, the result is invariant to the selector: the adaptive elastic net (Zou and Zhang 2009), the relaxed lasso, a random-forest permutation screen, and a best subset constrained to one factor per risk category all leave the alpha intact under honest split-sample inference (factors chosen on the first half, alpha estimated on the second). Most conservative of all, model-X knockoffs (Candès et al. 2018) at a 10% false discovery rate select *no factor at all*: at a controlled error rate the data neither identify a stable factor set nor produce one that absorbs the premium. Third, the alpha is not constant. Splitting on the same stress regimes used for costs (Figure 1, Panel a), the best-subset alpha rises monotonically for all three strategies, most steeply for the CDS–bond basis, from $+2.7\%$ in LOW to $+7.6\%$ in HIGH.

That gradient is a clue, not an explanation. A premium that triples when intermediary balance sheets are strained is what a capital-constraint mechanism predicts—but it is also what a common, countercyclical risk factor would produce. Section 4 separates them.

4 The Holder-Base Experiment

Duffie (2010) argues that when intermediary balance-sheet capacity is the binding constraint on arbitrage, mispricings serviced by a common pool of capital co-move, the co-movement intensifies as the pool is depleted, transmission runs down a liquidity pecking order as dealers adjust their most liquid positions first (Mitchell and Pulvino 2012), and persistence tracks how much balance sheet each trade consumes. Brunnermeier and Pedersen (2009) supply the mechanism: funding and market liquidity reinforce each other in a spiral that forces simultaneous deleveraging across books.

We work with mispricing *magnitudes* $m_{i,t}$ rather than returns, because it is the distance of the basis from its no-arbitrage value that measures how far capital is failing to close the gap; $m_{i,t}$ is the cross-sectional mean absolute basis for each market, sign-adjusted so that $\Delta m > 0$ is a widening, aggregated monthly.

The identification rests on one structural fact. All three arbitrages are run by the same leveraged institutional arbitrageur—the BTP Italia basis is traded by institutions like the other two, so it is fully part of the experiment—but only two of the three underlying instruments sit on levered balance sheets. The BTP Italia is held by preferred-habitat investors in the sense of Vayanos and Vila (2021): households that use no leverage, face no margin calls, and do not unwind when dealer capital withdraws. If a common capital pool is the mechanism, the two institutional mispricings should co-move and tighten their link in stress, while the retail-held basis sits outside the pool and, if forced institutional selling coincides with a flight of household demand toward the linker, moves the other way. A common macroeconomic shock cannot generate that sign structure: it would move all three claims alike, whoever holds them.

Figure 1, Panel (b), and Table 3, Panel A, show that structure. The institutional pair correlates at $+0.19$

Table 3: The holder-base experiment. Panel A: Pearson correlations of monthly mispricing changes Δm, unconditional (Newey–West t-statistics) and within stress regime. The LOW-to-HIGH rise for the institutional pair is significant at 1% on a Fisher z test ($p = 0.005$) and at 10% after the Forbes and Rigobon (2002) variance correction ($p = 0.070$). Panel B: OLS of the first principal component of the three Δm series on intermediary balance-sheet proxies (one per Duffie channel) and, separately, on a macroeconomic placebo set; 149 and 151 common months, Newey–West HAC standard errors. ***,**,*: 1%, 5%, 10%.

Panel A: Co-movement of mispricing changes

Pair	Uncond.	LOW	MEDIUM	HIGH
CDS–Bond vs. iTraxx *(institutional pair)*	$+0.19^{*}$	-0.21^{*}	$+0.20^{*}$	$+0.32^{**}$
CDS–Bond vs. BTP Italia *(across bases)*	-0.30^{***}	-0.13	-0.32^{***}	-0.47^{**}
BTP Italia vs. iTraxx *(across bases)*	-0.10	$+0.15$	-0.15	-0.14

Panel B: What drives the common factor?

	Coefficient	t	
Intermediary balance sheet			$\bar{R}^2 = 0.233$
He et al. (2017) capital ratio	-4.28^{**}	-2.28	capital scarcer ⇒ wider
Libor–OIS spread	$+3.35^{***}$	$+3.28$	funding dearer ⇒ wider
Prime-broker CDS (5Y, EU)	$+0.35^{**}$	$+2.01$	dealer credit worse ⇒ wider
Market illiquidity	$+1.77^{***}$	$+2.83$	liquidity worse ⇒ wider
Macroeconomic placebo			$\bar{R}^2 = 0.106$
Macro uncertainty Δ	$+0.36$	$+0.03$	
Real-activity uncertainty Δ	$+10.31$	$+1.23$	
5y5y inflation expectations	-0.63	-0.66	
Economic policy uncertainty	$+0.003$	$+1.32$	

unconditionally and the correlation rises monotonically with stress, from -0.21 in LOW to $+0.32$ in HIGH—an increase significant at the 1% level on a Fisher z test ($p = 0.005$) and still at the 10% level after the Forbes and Rigobon (2002) correction for stress-driven volatility ($p = 0.070$), so it is a genuine tightening of the cross-market link rather than a variance artifact. The pairs that cross holder bases behave as the experiment predicts: the CDS–bond versus BTP Italia correlation is -0.30 on the full sample and deepens to -0.47 in high stress, and an interaction regression $\Delta m_{i,t} = \alpha + \beta_0 \Delta m_{j,t} + \beta_1(\Delta m_{j,t} \times z_t) + \gamma z_t + \varepsilon_t$, in which the level term γz_t absorbs the common widening so that β_1 isolates amplification, confirms the asymmetry: stress *amplifies* transmission within the institutional pair ($\beta_1 = +0.08$, 5% level) and *deepens the negative* link toward the retail-held basis ($\beta_1 = -0.64$, 1% level). Bivariate DCC-GARCH estimates show this is no artifact of averaging within regimes: the BTP–CDS conditional correlation is negative in every single month of the sample (Appendix D).

Timing and persistence complete the mechanism. Granger tests on a VAR of the three Δm series recover the liquidity pecking order: shocks originate in the most liquid, standardized market—the iTraxx index, where dealer inventory adjusts fastest—and propagate within a month to the cash CDS–bond basis ($p = 0.011$) and the BTP Italia basis ($p = 0.007$), never the reverse, and both links survive at two and three lags. Generalized impulse responses (Pesaran and Shin 1998), invariant to VAR ordering, trace the same liquid-to-illiquid path with the same signs: positive into the CDS–bond basis, negative into BTP Italia. Persistence lines up with funding dependence, the one margin on which the three trades differ mechanically. The CDS–bond basis is the only one that must finance a cash bond on a dealer balance sheet for the life of the position; its AR(1) mispricing half-life is 3.0 months against 2.2 for both the BTP Italia and the iTraxx skew—one-third longer than the two trades that need no such financing, and those two are statistically indistinguishable from each other in persistence, as the funding criterion (and not the liquidity ordering, at whose opposite ends they sit) implies.

Panel B closes the loop on the source of the co-movement. Regressed on one proxy per Duffie channel, the common factor behind the three mispricings loads on all four with the sign the theory assigns—a fall in intermediary capital, a rise in funding cost, a deterioration in dealer credit, and a rise in market illiquidity each predict wider mispricings—explaining 23.3% of its variation. The placebo set reverses the picture: on macroeconomic and real-activity uncertainty, long-term inflation expectations, and economic policy uncertainty, all four coefficients are insignificant and $\bar{R}^2$ falls by more than half. A factor built on macroeconomic fundamentals would have produced the opposite ranking.

Why, then, is the premium not competed away? The limits differ by trade but share one root: the post-crisis cost of intermediary balance sheet. The CDS–bond trade consumes repo capacity in a thin corporate-repo market subject to recall risk (Bai and Collin-Dufresne 2019; Mitchell and Pulvino 2012). The iTraxx skew consumes regulatory capital—Boyarchenko et al. (2016) document that as required leverage ratios rose from 2–3% to 5–6%, the return on equity of index-versus-constituent trades fell by a factor of two to three, producing what they call "new normal" basis levels. And the BTP Italia basis rests on a retail holder base that neither monitors nor arbitrages it (Vayanos and Vila 2021). The alpha of Section 3 is not an anomaly awaiting correction; it is the equilibrium rent on a permanently scarcer balance sheet.

5 Implications for Investors

The results translate into four practical statements, and each rests on a specific number above rather than on the general proposition that arbitrage pays.

Size the book to the funding cycle, not to the average. The premium is state-dependent by roughly a factor of three between calm and stressed markets (Figure 1, Panel a), and it is measured net of transaction costs that we deliberately escalate in exactly those states. The implication runs against instinct: the correct response to a widening basis is to add, not to cut, because widening *is* the compensation mechanism. The constraint on doing so is not conviction but balance sheet, which is why the trade remains available—an investor with committed, unlevered capital is being paid for supplying precisely the resource that is scarce. The stress regimes here are observable in real time from a single quoted number, the iTraxx Main 5-year spread, so the rule is implementable rather than retrospective.

The retail-held basis is an internal hedge, not a third correlated bet. A book holding all three trades is not three units of the same exposure. The CDS–bond and iTraxx legs correlate at $+0.32$ in stress, but the BTP Italia leg correlates at -0.47 with the CDS–bond basis in exactly those months, and DCC estimates put that correlation below zero in every month of the sample. This is the diversification that fails to arrive in most crisis-correlated books, and it arrives here for a structural reason—the holder base cannot be forced to sell—rather than because of a historical correlation that may not repeat. The practical consequence is that the marginal risk contribution of the BTP Italia leg to a combined book is negative in stress, which argues for sizing it against the credit book rather than in isolation.

The persistence is structural, so the horizon must match it. These are not fast convergence trades. Mispricing half-lives are two to three months, average holding periods run from seven months (iTraxx) to over a year (BTP Italia), and 91–97% of trades exit at target rather than at maturity or sample end. Capital committed here must tolerate mark-to-market drawdown before convergence—which is the whole point, since it is that tolerance that earns the premium. The evidence that the limits are structural is what makes the horizon defensible: post-crisis leverage-ratio requirements and a thin corporate repo market are not features that mean-revert on a fund's reporting cycle.

Monitor the mechanism, not the level. Because the iTraxx skew Granger-causes the other two mispricings and never the reverse, the most liquid of the three is a leading indicator for the other two—dealer inventory adjusts there first. And because the common factor loads on Libor–OIS, dealer CDS, and intermediary capital rather than on macro data, the state variables worth watching are balance-sheet prices, not the macro calendar. The corresponding risk is symmetric: the same evidence that makes the premium durable identifies what would end it. A structural expansion of dealer balance-sheet capacity, or an institutionalization of the BTP Italia holder base, would compress these bases the way institutional entry compressed the TIPS–Treasury basis after 2009 (Fleckenstein et al. 2014). Those are observable conditions, and they are the ones to track.

6 Conclusion

Three European fixed-income arbitrages—each documented as a pricing anomaly, none previously traded systematically—earn Sharpe ratios of 0.98 to 1.56 net of costs, with alpha of 2.2% to 4.8% per year that no layer of factor controls absorbs: not three canonical benchmarks, not eight principal components, not a best-subset search over 75 tradable candidates, in sample or out. The premium concentrates in funding stress, and its cross-sectional behavior carries the signature of slow-moving capital: the two institutionally held mispricings co-move ever more tightly as intermediary capital tightens, while the retail-held BTP Italia basis moves against them—a sign structure a common risk factor cannot produce, and which identifies the balance-sheet channel of Duffie (2010). For investors the package is a state-dependent premium with an internal hedge, kept open by post-crisis regulation rather than by risk. The geography of arbitrage capital, not the asset-class label, determines what co-moves with what—and, for those able to supply balance sheet when it is scarce, what still pays.

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A Strategy Construction and Additional Performance

Every trade opens when the basis crosses an entry threshold and closes on reversion through an exit threshold, on maturity of the underlying, or at the end of the sample. All strategies require a minimum of six months to maturity at entry; trades shorter than three trading days are excluded ex post. The CDS–bond universe is restricted to single-name CDS whose reference entity belongs to the current on-the-run iTraxx Main series, which supplies both a liquidity filter and a transparent investment-grade screen; names that leave the index initiate no new trades, while open positions run to convergence or maturity. All CDS are centrally cleared at LCH or ICE, so the protection leg carries no direct counterparty exposure.

Table A.I: Entry and exit thresholds by strategy. All thresholds in bps. The CDS–bond basis is a one-directional negative basis trade (long bond, buy protection); the reverse trade is excluded because it requires a corporate reverse repo subject to recall risk in stress. iTraxx thresholds are calibrated to the historical volatility of each sub-index skew.

Strategy	Entry long	Entry short	Exit long	Exit short
BTP Italia	> 40	< -50	< 10	> -10
CDS–Bond Basis	< -40	–	> 0	–
iTraxx Main	> 5	< -10	< -5	> 10
iTraxx SnrFin	> 8	< -15	< -5	> 5
iTraxx SubFin	> 10	< -20	< -10	> 10
iTraxx Xover	> 15	< -25	< -15	> 10

Table A.II: Trade-level statistics. P&L is the cumulative return per trade in percentage points; duration in calendar days. Exit reasons may not sum to 100% due to rounding.

	BTP Italia	iTraxx Skew	CDS–Bond Basis
Total trades	260	206	931
Long	224	146	931
Short	36	60	0
Avg. duration (days)	386	225	240
Median duration (days)	212	97	119
Avg. P&L per trade (%)	2.079	1.491	2.463
Median P&L per trade (%)	1.960	1.395	1.927
Exit reasons (%)			
Target hit	92.3	97.1	90.7
Maturity	6.9	0.5	1.0
End of sample	0.8	2.4	8.4

Table A.III: Transaction fee schedule (bps per unit DV01 or duration). The fee is applied as entry DV01 $\times$ fee at entry plus exit DV01 $\times$ fee at exit; the exit fee is waived for trades held to maturity. The regime is set by the iTraxx Main 5Y level at the trade date. Because fees rise monotonically into stress, cost assumptions are most conservative exactly where gross arbitrage returns are largest.

	LOW (Main $<$ 60)	MEDIUM (60–100)	HIGH (Main $>$ 100)
BTP Italia	3.0	5.0	8.0
CDS–Bond Basis	4.0	7.0	10.0
iTraxx Main	0.5	0.75	1.0
iTraxx SnrFin	0.5	0.75	1.0
iTraxx SubFin	1.0	2.0	3.0
iTraxx Xover	3.0	5.0	7.0

Table A.IV: Equal-weighted strategy returns. Equal-weighted counterpart of Table 1, in which each active trade receives weight $1/K_t$ (Duarte et al. 2007) rather than weight proportional to the entry dislocation. Signal weighting dominates on Sharpe and MPPM for all three strategies, but every conclusion in the report survives the alternative scheme.

Strategy	n	Mean (% p.a.)	t	Std. Dev. (% p.a.)	Skew	Kurt	%Neg	Sharpe
BTP Italia	163	1.34	3.01	1.69	0.82	1.88	39.9	0.79
CDS–Bond Basis	207	3.84	5.71	2.62	0.49	5.28	28.0	1.46
iTraxx Skew	243	1.37	5.30	1.05	0.26	3.68	39.9	1.30

Table A.V: Manipulation-proof performance measure by risk aversion (Goetzmann et al. 2007). Computed from monthly returns net of costs; the risk-free rate is 1-month Euribor. The near-invariance of the MPPM to ρ shows that risk-adjusted performance does not rely on tail events: an investor with $\rho = 4$ evaluates these strategies almost exactly as one with $\rho = 2$.

	Signal-weighted			Equal-weighted		
Strategy	$\rho = 2$	$\rho = 3$	$\rho = 4$	$\rho = 2$	$\rho = 3$	$\rho = 4$
BTP Italia	1.50	1.48	1.47	1.24	1.22	1.21
CDS–Bond Basis	4.43	4.39	4.35	3.66	3.63	3.60
iTraxx Skew	1.88	1.87	1.86	1.35	1.34	1.34

Table A.VI: Volatility-managed portfolios (Moreira and Muir 2017). The managed portfolio scales exposure inversely to the previous month's realized variance, $f^{\sigma}_{t+1} = (c/\hat{\sigma}^2_t) f_{t+1}$, with c matching buy-and-hold volatility. Alpha is the intercept from regressing managed on buy-and-hold returns, Newey–West (4 lags). A significant alpha diagnoses a time-varying μ_t/σ^2_t. It appears only for the one strategy that finances a cash bond—consistent with a funding-liquidity mechanism, and the first indication of the funding asymmetry that Section 4 tests directly.

	Buy-and-hold			Vol-managed				
Strategy	Ret	Vol	Sharpe	Ret	Vol	Sharpe	α	$t(\alpha)$
Panel A: Signal-weighted								
BTP Italia	1.52	1.60	0.95	0.37	1.60	0.23	−0.22	−0.50
CDS–Bond Basis	4.52	2.88	1.57	3.08	2.88	1.07	+1.39*	+1.90
iTraxx Skew	1.91	1.53	1.25	0.84	1.53	0.55	−0.16	−0.58
Panel B: Equal-weighted								
BTP Italia	1.27	1.67	0.76	0.21	1.67	0.13	−0.33	−0.73
CDS–Bond Basis	3.73	2.53	1.48	2.45	2.53	0.97	+0.93	+1.38
iTraxx Skew	1.36	1.05	1.29	0.60	1.05	0.57	−0.01	−0.09

B Benchmark Factor Models

Full euro-panel estimates of the three benchmark specifications summarized in Panel A of Table 2, followed by the consolidated alpha synthesis (unconditional, conditional, and out-of-sample). Inference throughout uses Newey–West HAC standard errors with the Newey and West (1994) truncation lag. US-factor panels, sub-period splits, rolling-window out-of-sample counterparts, and VIF diagnostics are available in the underlying dissertation chapter.

Table A.VII: Duarte et al. (2007) factor model. $r_{i,t} = \alpha_i + \beta_1 \text{Mkt}_t + \beta_2 \text{SMB}_t + \beta_3 \text{HML}_t + \beta_4 \text{UMD}_t + \beta_5 R_{S,t} + \beta_6 R_{I,t} + \beta_7 R_{B,t} + \beta_8 R_{2,t} + \beta_9 R_{5,t} + \beta_{10} R_{10,t} + \varepsilon_{i,t}$, euro-area counterparts. R_S: EuroStoxx Banks excess return. R_I, R_B: European A-rated industrial and bank bond indices. R_2, R_5, R_{10}: European 2-, 5-, and 10-year government bond portfolios. α annualized.

	α	β									
	(% p.a.)	Mkt	SMB	HML	UMD	R_S	R_I	R_B	R_2	R_5	R_{10}
BTP Italia	1.62***	−0.04*	−0.06***	−0.03	0.02	0.01	−0.03	0.04	0.03	−0.04	−0.02
	(3.29)	(−1.79)	(−2.86)	(−1.42)	(1.13)	(1.01)	(−0.27)	(0.49)	(0.06)	(−0.14)	(−0.16)
CDS–Bond	4.24***	0.00	0.04	0.03	−0.02	−0.02*	0.45***	−0.06	−0.27	−0.04	−0.10
	(5.27)	(0.20)	(0.90)	(1.11)	(−0.88)	(−1.74)	(3.94)	(−1.57)	(−0.45)	(−0.10)	(−0.71)
iTraxx Skew	2.19***	−0.02	−0.03**	−0.07**	−0.02*	0.01	0.05	−0.02	0.06	0.05	−0.06
	(4.90)	(−1.25)	(−2.04)	(−2.26)	(−1.95)	(1.53)	(0.75)	(−0.78)	(0.16)	(0.21)	(−0.78)
$\bar{R}^2$	BTP Italia 0.11	CDS–Bond 0.15	iTraxx Skew 0.06								

Table A.VIII: Fung and Hsieh (2004) factor model. Seven-factor asset-based style model, euro-area counterparts. BdOpt, FXOpt, ComOpt are lookback-straddle trend-following returns on government-bond, currency, and commodity futures. 10Y: excess return on a 10-year Bund total-return index. CredSpr: euro-area BBB corporates less 10-year Bunds, in total-return form. α annualized.

	α	β						
	(% p.a.)	S&P	SC–LC	BdOpt	FXOpt	ComOpt	10Y	CredSpr
BTP Italia	1.81***	−0.03**	−0.06***	0.00	0.00	0.00	−0.01	0.02
	(4.08)	(−2.53)	(−2.70)	(1.25)	(0.03)	(1.09)	(−0.27)	(0.30)
CDS–Bond	3.98***	−0.02	0.05	0.00	−0.01*	−0.00	0.17***	0.19***
	(5.05)	(−1.18)	(1.24)	(1.01)	(−1.95)	(−0.42)	(3.43)	(2.87)
iTraxx Skew	1.84***	0.00	−0.02	0.00	0.00	0.00	0.03	0.02
	(4.49)	(0.41)	(−1.50)	(0.60)	(1.01)	(0.57)	(0.83)	(0.62)
$\bar{R}^2$	BTP Italia 0.10	CDS–Bond 0.11	iTraxx Skew 0.00					

Table A.IX: Brooks et al. (2020) active fixed-income factor model. Eight traditional risk premia organized along duration, credit, and volatility-selling channels, euro-area counterparts. RatesVol is the excess return on a one-month variance swap on the ten-year Bund future. α annualized.

	α	β							
	(% p.a.)	Term	GlTerm	GlAgg	InflLink	CorpCred	EmDebt	EmCurr	RatesVol
BTP Italia	2.12***	−0.01	−0.05	0.13	−0.06	−0.12**	0.05	−0.01	−0.16
	(3.84)	(−0.47)	(−0.16)	(0.49)	(−1.60)	(−2.23)	(0.97)	(−0.30)	(−1.36)
CDS–Bond	4.20***	0.08*	−0.43	0.30	−0.07	−0.03	0.14***	0.01	0.32
	(5.20)	(1.89)	(−0.73)	(0.56)	(−1.05)	(−0.53)	(3.15)	(0.16)	(1.15)
iTraxx Skew	1.75***	−0.04*	−0.09	0.33	−0.00	0.02	−0.08**	0.01	0.19
	(4.24)	(−1.92)	(−0.42)	(1.42)	(−0.01)	(1.61)	(−2.18)	(0.55)	(1.53)
$\bar{R}^2$	BTP Italia 0.08	CDS–Bond 0.12	iTraxx Skew 0.04						

Table A.X: Alpha across benchmark models: unconditional, conditional, and out-of-sample. Panel A repeats the full-sample intercepts. Panel B reports the Christopherson et al. (1998) full conditional model $r_t = \alpha_0 + \alpha_1 z_{t-1} + (\beta + \delta z_{t-1})' X_t + \varepsilon_t$, in which both the alpha and the loadings are conditioned on z_{t-1}, the one-month-lagged standardized iTraxx Main 5Y spread—a *predetermined* conditioning variable, so α_1 measures conditional performance rather than contemporaneous co-movement. Significance of α_1 from a moving-block bootstrap ($B = 9{,}999$, $H_0 : \alpha_1 = 0$ imposed, block length 9). Panel C reports out-of-sample alpha and information ratio in the Welch and Goyal (2008) recursive design; N is the number of out-of-sample months, common across benchmarks within each strategy.

	Duarte et al. (2007)			Brooks et al. (2020)			Fung and Hsieh (2004)		
Panel A: Full-sample alpha (% p.a.)									
	α	t	N	α	t	N	α	t	N
BTP Italia	$+1.62^{***}$	3.29	163	$+2.12^{***}$	3.84	163	$+1.81^{***}$	4.08	163
iTraxx Skew	$+2.19^{***}$	4.90	243	$+1.75^{***}$	4.24	243	$+1.84^{***}$	4.49	243
CDS–Bond Basis	$+4.24^{***}$	5.27	207	$+4.20^{***}$	5.20	207	$+3.98^{***}$	5.05	207
Panel B: Christopherson et al. (1998) conditional alpha (% p.a.)									
	α_0	α_1	$\bar{R}^2$	α_0	α_1	$\bar{R}^2$	α_0	α_1	$\bar{R}^2$
BTP Italia	$+1.58^{***}$	$+1.20^{**}$	0.23	$+1.91^{***}$	$+1.02$	0.13	$+1.70^{***}$	$+0.50$	0.19
	(3.78)	(2.25)		(3.03)	(1.14)		(5.24)	(1.22)	
iTraxx Skew	$+1.94^{***}$	$+0.93^{**}$	0.11	$+1.82^{***}$	$+1.43^{***}$	0.14	$+1.89^{***}$	$+1.51^{***}$	0.05
	(4.75)	(2.55)		(4.43)	(3.40)		(4.85)	(4.66)	
CDS–Bond Basis	$+4.67^{***}$	$+2.10^{**}$	0.25	$+4.27^{***}$	$+3.18^{***}$	0.18	$+3.96^{***}$	$+1.73^{**}$	0.17
	(5.99)	(2.74)		(6.24)	(4.02)		(5.45)	(2.47)	
Panel C: Out-of-sample alpha (% p.a.), expanding window									
	α	t	IR	α	t	IR	α	t	IR
BTP Italia ($N = 103$)	$+1.29^{**}$	2.24	$+0.78$	$+2.10^{***}$	3.83	$+1.25$	$+1.69^{***}$	2.86	$+1.05$
iTraxx Skew ($N = 183$)	$+2.27^{***}$	4.71	$+1.37$	$+1.77^{***}$	3.48	$+1.07$	$+2.02^{***}$	4.05	$+1.20$
CDS–Bond ($N = 147$)	$+3.48^{***}$	4.29	$+1.25$	$+3.86^{***}$	4.30	$+1.31$	$+3.45^{***}$	3.87	$+1.19$

C Factor Universe and Selection Robustness

The candidate universe comprises 75 tradable risk factors drawn from the empirical asset pricing, intermediary, and macro-finance literatures, organized into five economic categories: credit risk, liquidity and funding, equity, volatility, and interest rates and term structure. Every candidate is the excess return on an implementable position—a cash index, a long-short factor portfolio, a derivative overlay, or a DV01-neutral curve trade. State variables enter through traded counterparts rather than as raw proxies: the intermediary capital factor of He et al. (2017) enters through their tradable primary-dealer equity factor, which tracks innovations in the capital ratio with a correlation of 0.92, and aggregate liquidity through the tradable factor of Pástor and Stambaugh (2003). Dollar-denominated factors enter as euro excess returns, converted following Glück et al. (2021). Because every regressor is a realized excess return, each intercept retains its Jensen-alpha interpretation.

For selection, six candidates that are near-duplicates of another factor (absolute correlation above 0.95) are dropped, leaving $p = 69$; each factor is standardized to unit L_2 norm so that which factors enter is scale-invariant, with point estimates and inference computed on original units. At $p = 69$ the subset space ($2^{69} \approx 6 \times 10^{20}$) rules out exhaustive search, so the ℓ_0 problem $\min_{\alpha,\beta} T^{-1} \sum_t (r_t - \alpha - \beta' f_t)^2$ s.t. $\|\beta\|_0 \leq k$ is solved by the splicing algorithm of Zhu et al. (2020), polished against a greedy forward search. Cardinality is fixed at $k = 8$ for parsimony and cross-strategy comparability rather than chosen by cross-validation, whose objective is nearly flat—and whose argmin therefore ill-determined—on alpha-dominated series with weak factor signal.

Table A.XI: PCA spanning regressions and robustness to the number of components. Full-sample contemporaneous spanning $r_{i,t} = \alpha_i + \beta_i' \text{PC}_t + \varepsilon_{i,t}$ on the first K principal components of the standardized candidate panel; the baseline $K = 8$ is selected by the Bai and Ng (2002) IC_{p2} criterion. Expl. Var. is cumulative variance captured. The alpha is flat in K: adding components raises explanatory power without touching the mean.

	BTP Italia		CDS–Bond Basis		iTraxx Skew		Expl. Var.
K	α	$\bar{R}^2$	α	$\bar{R}^2$	α	$\bar{R}^2$	(%)
5	1.50*** (3.23)	0.089	4.79*** (6.22)	0.114	2.21*** (5.39)	0.050	55.0
8	1.44*** (3.30)	0.088	4.78*** (6.25)	0.109	2.21*** (5.95)	0.134	64.7
11	1.47*** (3.56)	0.086	4.77*** (6.13)	0.111	2.21*** (5.95)	0.136	71.8
13	1.75*** (4.05)	0.144	4.76*** (6.23)	0.105	2.21*** (6.12)	0.146	75.6

Table A.XII: Top factor loadings by principal component. For each of the first three components, the six candidates with the largest absolute eigenvector entry. The components admit a clear economic reading—PC1 credit and equity, PC2 global rates and bonds, PC3 swap spreads and dealer CDS—confirming that the candidate panel spans the intended risk space even though its diffuse combinations do not span strategy returns.

PC1		PC2		PC3	
Factor	Loading	Factor	Loading	Factor	Loading
CRED_SPR_EU	+0.20	GLOBAL_AGG	+0.30	SNRFIN_MAIN	+0.30
DEF_US	+0.20	GLOBAL_TERM	+0.28	BTP_BUND_2Y	+0.21
DEF_EU	+0.19	R10_EU	+0.28	PB_CDS_5Y_EU	+0.20
CRED_SPR_US	+0.19	GOVT_EU	+0.28	CREDIT_US	−0.20
ITRX_MAIN	+0.19	TERM_EU	+0.28	BTP_BUND	+0.19
HY_CORP	+0.19	RI_EU	+0.27	FI_VAL	+0.19

Table A.XIII: Post-selection OLS on the best-subset factors. $r_{i,t} = \alpha_i + \sum_j \beta_{i,j} F_{j,t} + \varepsilon_{i,t}$ on the factors selected for that strategy (Bertsimas et al. 2016), in original units. The 95% CI is the percentile interval for α from a stationary bootstrap (Politis and Romano 1994), 9,999 replications, expected block length nine months. The three sets are pairwise disjoint.

	Coefficient	t	95% CI
Panel A: BTP Italia	$T = 157$, $k = 8$, $\bar{R}^2 = 0.31$		
α (% p.a.)	2.2^{***}	5.59	[1.3, 3.0]
DEF_EU	0.058^{**}	2.43	
LIQ_V	-0.016^{*}	-1.88	
MKT_EU	-0.038^{***}	-3.99	
SMB_EU	-0.056^{***}	-2.79	
FI_VAL	-0.094^{**}	-2.28	
FI_DEF	-0.207^{***}	-4.94	
5Y5Y_INFL_US	-0.232^{***}	-2.70	
SS5Y	-0.010^{**}	-2.26	
Panel B: CDS–Bond Basis	$T = 197$, $k = 8$, $\bar{R}^2 = 0.21$		
α (% p.a.)	4.8^{***}	5.87	[2.9, 6.6]
PB_CDS_5Y_EU	-0.223^{**}	-2.28	
DEF_US	0.112^{***}	3.21	
HY_CORP	-0.109^{**}	-2.20	
EMERG_DEBT	0.074^{**}	2.17	
RI_EU	0.217^{***}	3.06	
FX_MOM	-0.041	-1.33	
PTFSFX	-0.006^{**}	-2.26	
CURV_2S5S10S_EU	-0.015^{*}	-1.82	
Panel C: iTraxx Skew	$T = 209$, $k = 8$, $\bar{R}^2 = 0.25$		
α (% p.a.)	2.9^{***}	5.30	[1.5, 3.8]
CDX_IG	0.167^{**}	2.40	
HKM_IC	-0.025^{***}	-4.14	
HML_EU	-0.026^{*}	-1.91	
BAB_EU	-0.044^{***}	-4.26	
GMOM	-0.052^{***}	-3.05	
FI_MOM	-0.041^{***}	-3.87	
SVS_RET_1M	-0.857^{**}	-2.52	
ATM_IV_ITRX	0.304	1.32	

Table A.XIV: Alpha across factor-selection methods. n is the number of selected factors. α^{split} is the honest split-sample alpha—factors selected on the first half of the sample only, alpha estimated as the intercept of an OLS on the second half, which is valid post-selection inference by sample splitting. α^{OOS} is the frozen-set out-of-sample alpha. Selection with zero factors reports the raw strategy alpha. The knockoffs row is the sharpest statement of the two-sided message: at a controlled false discovery rate, no candidate carries enough signal to be selected at all, so the data neither identify a stable factor set nor produce one that absorbs the premium.

	BTP Italia			CDS–Bond Basis			iTraxx Skew		
Method	n	α^{split}	α^{OOS}	n	α^{split}	α^{OOS}	n	α^{split}	α^{OOS}
Best subset (ℓ_0, $k=8$)	8	$+1.69^{***}$	$+2.05^{***}$	8	$+2.92^{***}$	$+3.90^{***}$	8	$+2.49^{***}$	$+2.50^{***}$
		(2.67)	(3.96)		(4.31)	(4.43)		(3.59)	(5.35)
Best subset, one per category	5	$+1.91^{***}$	$+1.23^{***}$	5	$+2.52^{***}$	$+3.21^{***}$	5	$+2.18^{***}$	$+2.18^{***}$
		(2.83)	(2.59)		(3.47)	(3.58)		(3.19)	(4.47)
Adaptive elastic net	4	$+1.96^{***}$	$+1.43^{***}$	4	$+2.85^{***}$	$+3.29^{***}$	21	$+2.10^{***}$	$+2.74^{***}$
		(2.94)	(2.91)		(4.40)	(3.70)		(4.30)	(6.50)
Relaxed lasso	12	$+1.96^{***}$	$+1.60^{***}$	2	$+3.11^{***}$	$+3.54^{***}$	0	$+2.20^{***}$	$+1.79^{***}$
		(2.94)	(3.02)		(3.76)	(3.85)		(3.00)	(3.36)
Random-forest screen (top 8)	8	$+1.54^{***}$	$+1.30^{***}$	8	$+3.41^{***}$	$+3.80^{***}$	8	$+2.19^{***}$	$+1.87^{***}$
		(3.44)	(2.73)		(4.03)	(4.29)		(3.76)	(3.66)
Model-X knockoffs (FDR 10%)	0	$+1.96^{***}$	$+1.79^{***}$	0	$+3.11^{***}$	$+3.77^{***}$	0	$+2.06^{***}$	$+1.79^{***}$
		(2.94)	(3.19)		(3.76)	(4.04)		(2.81)	(3.36)

Table A.XV: Best-subset support as the cardinality k varies. The identity of the selected factors shifts with k, as expected when correlated candidates substitute for one another; the alpha does not. The post-double-selection control set of Belloni et al. (2014)—which selects controls both for the strategy return and for the factors of interest—is a strict subset of the consensus exposures (FI_DEF for BTP Italia; EMERG_DEBT and RI_EU for the CDS–bond basis; empty for the iTraxx skew).

k	α (% p.a.)	$\bar{R}^2$	Selected factors
BTP Italia			
5	$+2.24^{***}$ (5.84)	0.27	MKT_EU, SMB_EU, FI_VAL, FI_DEF, SS5Y
8	$+2.19^{***}$ (5.59)	0.31	DEF_EU, LIQ_V, MKT_EU, SMB_EU, FI_VAL, FI_DEF, 5Y5Y_INFL_US, SS5Y
10	$+1.85^{***}$ (4.07)	0.32	DEF_EU, LIQ_V, MKT_EU, SMB_EU, UMD_EU, FI_VAL, FI_DEF, PTFSBD, 5Y5Y_INFL_US, SS5Y
CDS–Bond Basis			
5	$+4.67^{***}$ (5.76)	0.18	PB_CDS_1Y_EU, DEF_US, EMERG_DEBT, RI_EU, PTFSFX
8	$+4.80^{***}$ (5.87)	0.21	PB_CDS_5Y_EU, DEF_US, HY_CORP, EMERG_DEBT, RI_EU, FX_MOM, PTFSFX, CURV_2S5S10S_EU
10	$+3.87^{***}$ (5.32)	0.22	EMERG_DEBT, RI_EU, SMB_EU, COM_MOM, PTFSFX, SVS_RET_1M, IV_BUND, INFL_LINK, CURV_2S5S10S_EU, SS5Y
iTraxx Skew			
5	$+3.02^{***}$ (4.99)	0.19	HKM_IC, BAB_EU, FI_MOM, SVS_RET_1M, ATM_IV_ITRX
8	$+2.93^{***}$ (5.30)	0.25	CDX_IG, HKM_IC, HML_EU, BAB_EU, GMOM, FI_MOM, SVS_RET_1M, ATM_IV_ITRX
10	$+2.72^{***}$ (5.54)	0.28	CDX_IG, DEF_US, HKM_IC, HML_EU, BAB_EU, GMOM, FI_MOM, SVS_RET_1M, ATM_IV_ITRX, SLOPE_2S10S_US

Table A.XVI: Alpha by stress regime: PCA and best-subset controls. Annualized alpha per regime from a single regression with three intercept dummies and betas common across regimes (no constant), $r_t = \sum_g \alpha_g D_{g,t} + \sum_j \beta_j X_{jt} + \varepsilon_t$, Newey–West (4 lags). These are the estimates plotted in Panel (a) of Figure 1.

	PCA			Best-Subset		
Strategy	LOW	MEDIUM	HIGH	LOW	MEDIUM	HIGH
BTP Italia	$+0.37$	$+1.68^{***}$	$+3.27^{*}$	$+1.28^{**}$	$+2.40^{***}$	$+3.77^{***}$
	(0.65)	(3.17)	(1.82)	(2.29)	(5.08)	(2.95)
CDS–Bond Basis	$+2.87^{***}$	$+4.44^{***}$	$+7.55^{***}$	$+2.69^{***}$	$+4.69^{***}$	$+7.59^{***}$
	(3.12)	(3.87)	(4.84)	(2.99)	(3.95)	(4.93)
iTraxx Skew	$+1.03^{*}$	$+2.23^{***}$	$+3.45^{***}$	$+2.48^{***}$	$+2.94^{***}$	$+3.39^{***}$
	(1.95)	(4.62)	(4.32)	(3.95)	(4.38)	(4.14)

D Cross-Strategy Interdependencies

The complete slow-moving capital test battery behind Section 4: unconditional and regime-dependent correlations with their significance tests, the stress-amplification interaction regressions, cross-strategy spanning regressions, Granger causality across lag lengths, and AR(1) persistence.

Table A.XVII: Correlations of mispricing changes, unconditional and by regime. Unconditional columns are Pearson correlations with Newey–West HAC t-statistics (4 lags). Regime columns are within-regime correlations, with stars from two-sided t-tests. The HIGH_{FR} column applies the Forbes and Rigobon (2002) variance adjustment and tests the LOW-to-HIGH change by Fisher z-transformation, so it isolates a genuine tightening of the cross-market link from the mechanical rise in correlation that accompanies higher mispricing volatility in stress.

	Unconditional		By regime			
Pair	Pearson	t	LOW	MED	HIGH	HIGH_{FR}
CDS–Bond vs BTP Italia	-0.298^{***}	-3.04	-0.134	-0.324^{***}	-0.465^{**}	-0.326
CDS–Bond vs iTraxx Skew	$+0.191^{*}$	$+1.73$	-0.208^{*}	$+0.201^{*}$	$+0.316^{**}$	$+0.135^{*}$
BTP Italia vs iTraxx Skew	-0.100	-1.07	$+0.150$	-0.147	-0.144	-0.069

Table A.XVIII: Stress amplification: interaction regressions. $\Delta m_{i,t} = \alpha + \beta_0 \Delta m_{j,t} + \beta_1(\Delta m_{j,t} \times z_t) + \gamma z_t + \varepsilon_t$, where z_t is the contemporaneous standardized iTraxx Main 5Y level. The state enters contemporaneously because the object is amplification, not predetermined conditional performance; the level term γz_t absorbs the direct widening common to both series, so β_1 isolates amplification. Newey–West HAC (4 lags).

Dependent	Variable	Coefficient	t	p	$\bar{R}^2$
CDS–Bond Basis	Δm_j (BTP Italia)	-0.144^{**}	-2.02	0.043	0.1221
	$\Delta m_j \times z_t$	-0.086	-1.50	0.133	
	z_t (stress)	$+1.591^{***}$	$+2.63$	0.009	
BTP Italia	Δm_j (CDS–Bond Basis)	-0.464^{***}	-3.22	0.001	0.1286
	$\Delta m_j \times z_t$	-0.637^{***}	-2.79	0.005	
	z_t (stress)	$+1.502$	$+0.95$	0.341	
CDS–Bond Basis	Δm_j (iTraxx Skew)	$+0.482$	$+1.05$	0.291	0.0531
	$\Delta m_j \times z_t$	$+0.016$	$+0.07$	0.940	
	z_t (stress)	$+0.934^{**}$	$+2.29$	0.022	
iTraxx Skew	Δm_j (CDS–Bond Basis)	$+0.030$	$+1.13$	0.261	0.0731
	$\Delta m_j \times z_t$	$+0.079^{**}$	$+2.01$	0.044	
	z_t (stress)	-0.076	-0.31	0.757	
BTP Italia	Δm_j (iTraxx Skew)	-0.389	-0.66	0.510	0.0154
	$\Delta m_j \times z_t$	-0.618	-0.93	0.353	
	z_t (stress)	-0.128	-0.07	0.941	
iTraxx Skew	Δm_j (BTP Italia)	-0.010	-0.82	0.412	0.0201
	$\Delta m_j \times z_t$	-0.020	-1.01	0.313	
	z_t (stress)	-0.045	-0.19	0.849	

Table A.XIX: Cross-strategy spanning regressions in Δm. Each dependent variable is regressed on the contemporaneous value and the first two lags of the other two Δm series, in the design of Fleckenstein et al. (2014); the contemporaneous terms enter as controls so that the lagged loadings do not absorb same-month co-movement. Newey–West HAC (4 lags). Because the design omits each series' own lags, a predictive loading here cannot be distinguished from persistence in the predicted variable, which is why the Granger tests below embed the three series in a VAR that conditions on that history.

Dependent	Regressor	β	t	p	$\bar{R}^2$
CDS–Bond Basis	BTP Italia_L0	-0.148^{**}	-2.20	0.0282	0.0913
	BTP Italia_L1	-0.010	-0.22	0.8288	
	BTP Italia_L2	-0.039	-1.27	0.2034	
	iTraxx Skew_L0	$+0.370$	$+0.91$	0.3637	
	iTraxx Skew_L1	$+0.710^{*}$	$+1.93$	0.0530	
	iTraxx Skew_L2	$+0.030$	$+0.08$	0.9367	
BTP Italia	iTraxx Skew_L0	-0.687	-1.41	0.1581	0.1000
	iTraxx Skew_L1	-0.950	-1.36	0.1728	
	iTraxx Skew_L2	$+0.391$	$+0.46$	0.6437	
	CDS–Bond_L0	-0.424^{**}	-2.43	0.0150	
	CDS–Bond_L1	$+0.215$	$+1.57$	0.1161	
	CDS–Bond_L2	$+0.072$	$+0.62$	0.5340	
iTraxx Skew	BTP Italia_L0	-0.010	-0.76	0.4482	0.0103
	BTP Italia_L1	$+0.016$	$+0.80$	0.4210	
	BTP Italia_L2	$+0.028^{*}$	$+1.94$	0.0529	
	CDS–Bond_L0	$+0.017$	$+0.52$	0.5997	
	CDS–Bond_L1	$+0.019$	$+0.75$	0.4520	
	CDS–Bond_L2	$+0.008$	$+0.39$	0.6945	

Table A.XX: Granger causality: robustness to lag length. Each entry is the F-statistic for the null that the row variable does not Granger-cause the column variable, from a VAR(p) on the Δm series estimated separately at $p = 1, 2, 3$. Lag selection: HQC selects $p = 1$; AIC and FPE select $p = 2$; BIC selects $p = 0$ (forced to 1). The pecking order is one-directional at every lag: the liquid index leads both cash bases, and neither reverses.

		$p=1$		$p=2$		$p=3$	
Cause	Effect	F	p-val	F	p-val	F	p-val
BTP Italia	CDS–Bond Basis	0.05	0.8176	0.55	0.5790	1.61	0.1858
iTraxx Skew	CDS–Bond Basis	6.52^{**}	0.0110	3.97^{**}	0.0195	2.14^{*}	0.0946
CDS–Bond Basis	BTP Italia	3.08^{*}	0.0802	1.40	0.2465	1.53	0.2071
iTraxx Skew	BTP Italia	7.39^{***}	0.0068	4.16^{**}	0.0162	3.04^{**}	0.0289
CDS–Bond Basis	iTraxx Skew	0.01	0.9043	0.16	0.8537	0.13	0.9445
BTP Italia	iTraxx Skew	0.48	0.4897	1.22	0.2970	0.87	0.4589

Table A.XXI: AR(1) persistence of mispricing levels. $m_{i,t} = c_i + \phi_i m_{i,t-1} + u_{i,t}$, half-life $h_i^* = \ln(0.5)/\ln(\phi_i)$ months, Newey–West HAC (4 lags). Duffie (2010) makes the prediction quantitative: the half-life of the reversal that follows a shock is decreasing in the rate at which arbitrage capital reaches the trade, so half-lives should track funding frictions across strategies. They do, on the one distinction the three trades allow—the CDS–bond basis is the only one that must finance a cash bond, and it is a third more persistent than the two that need not, which are themselves indistinguishable from each other.

Strategy	ϕ	HAC t	Half-life (months)	N
CDS–Bond Basis	0.7913^{***}	14.43	3.0	148
BTP Italia	0.7272^{***}	13.54	2.2	148
iTraxx Skew	0.7258^{***}	9.06	2.2	148

E Data and Replication

Data sources: Borsa Italiana MOT (BTP Italia and nominal BTP prices), Bloomberg (zero-coupon inflation swap rates and corporate bond spreads), CMA/CME Group (single-name CDS spreads), and J.P. Morgan DataQuery (iTraxx index and constituent data, cross-validated against Barclays Live). Factor data are drawn from the Kenneth French data library, the AQR data library, the He et al. (2017) intermediary asset pricing archive, the Pástor and Stambaugh (2003) liquidity series, FRED, the ECB Statistical Data Warehouse, and the Office of Financial Research. All samples end in May 2025.

The replication pipeline is written in Python as a structured package: raw vendor pulls are cached to parquet, strategy construction and index aggregation are deterministic given the threshold table (Table A.I) and the fee schedule (Table A.III), and every table and figure in this report is regenerated from a single entry point. It is available from the author on request.

Limitations. Three are worth stating plainly. First, the samples are not of common length—163, 207, and 243 months—so the cross-strategy tests of Section 4 run on the 148 to 151 overlapping months, which is the binding constraint on their power. Second, transaction costs are modelled from a regime-dependent DV01 schedule calibrated to observed dealer quotes rather than from executed fills; the schedule is deliberately conservative in stress, but it is a model. Third, the identification in Section 4 rests on the BTP Italia holder base being materially more retail than the other two instruments, which is a structural feature of the Italian Treasury's placement mechanism rather than a randomized assignment.